

Attention and Green Delivery

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Abstract

This paper shows that attention shapes sustainable household choice during the green transition. Using transaction-level data from Singapore’s largest online grocery platform, we study the adoption of a green delivery option that reduces carbon emissions through route consolidation. We find that households are less likely to choose green delivery when temperatures are higher, while environment-related public events are associated with higher adoption. However, this event-day association is weaker under greater environmental stress. We interpret these findings through a behavioral inattention framework: heat stress shifts attention toward immediate comfort and convenience, reducing the salience of future environmental benefits, whereas public environmental cues temporarily redirect attention toward environmental goals.

Keywords: attention, cognitive scarcity, sustainability, green delivery

JEL Classification: D12, D83, D91, Q54, Q58

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1 Introduction

Household behavior is central to the green transition (Lyubich 2025; Kaufmann et al. 2024). Residential energy use accounts for a substantial share of global emissions, and household decisions over food, transport, and durable consumption further shape the carbon intensity of everyday economic activity (IPCC 2022). Yet many environmentally relevant choices involve trade-offs between social benefits and immediate private costs, such as price, convenience, time, or effort. Evidence from energy efficiency and durable-goods markets points to this friction: households often underweight future operating costs, underreact to information, and make choices that deviate from standard cost-benefit predictions (Allcott and Wozny 2014; Fowlie et al. 2018). These findings suggest that attention and salience may shape how households trade off private costs against social benefits in environmentally relevant decisions (DellaVigna 2009; Nordhaus 2019). A key question, then, is whether making environmental concerns more salient increases green behavior, or whether attention instead shifts toward more immediate private needs.

The direction of this effect is theoretically ambiguous because salience can shift attention in opposing ways. On the one hand, adverse environmental conditions may reduce green behavior by making immediate private concerns more pressing. Heat, humidity, or poor air quality can increase the salience of comfort, convenience, and effort costs, thereby reducing the weight placed on delayed environmental benefits. This mechanism builds on evidence that stress and cognitive scarcity can distort decision-making (Gabaix 2019; Mullainathan and Shafir 2013). On the other hand, public environmental cues may increase green behavior by drawing attention to the social benefits of green choices and making environmental considerations more prominent at the moment of decision. As earlier studies show, information, media coverage, and social norms can shift climate-related behavior by changing what individuals notice (Beattie 2025; Andre et al. 2024). Whether green behavior rises or falls thus depends on which dimension of salience dominates: environmental benefits or immediate private costs. Yet we have limited evidence on how these competing forces interact in everyday consumer decisions.

To address this gap, we use transaction-level data from RedMart, a major online grocer in Singapore, covering approximately 1,000,000 anonymized delivery orders between March 2023 and March 2025. This setting is well suited to our question because it provides a high-frequency revealed-preference measure of pro-environmental behavior at the moment of choice. Instead of relying on stated preferences or broad purchasing categories, we observe whether households select a lower-carbon delivery option in an everyday consumption decision. At checkout, customers choose among delivery slots that differ in timing, convenience, and, in some cases, environmental attributes. Some slots are labeled ‘Less CO₂’ because they are designed to enable route consolidation and reduce

delivery-related emissions. These labeled options carry no direct financial incentive. This allows us to study whether households trade off private convenience for an environmental benefit when the green attribute is made salient at checkout.

Specifically, we focus on orders in which at least one labeled slot is available and restrict attention to zero-fee slots to remove direct price confounds, yielding a working sample of about 325,000 orders. We then link these choices to two distinct forms of attention-relevant variation: public environmental cues and environmental stress. Public environmental cues come from environment-related public events that make environmental issues more visible. Environmental stress comes from day-to-day variation in heat and air pollution, which may either heighten environmental concern or redirect attention toward immediate private needs.

Singapore’s tropical climate generates meaningful variation in temperature and humidity, making it a useful setting for studying how environmental conditions shape household decisions. We focus on heat and air pollution because both are directly experienced in everyday life and may affect immediate discomfort as well as attention to environmental quality. Prior work also shows that both stressors can shape risk-related behavior, including insurance demand (Agarwal et al. 2025; Chang et al. 2018). To capture public environmental cues, we use environment-related public events, including highly publicized sustainability campaigns such as Clean & Green Singapore and Go Green SG.¹ Together, these features allow us to study how environmental stress and public environmental cues shape household decisions, and whether stress weakens the association between public events and green delivery choice at the moment of choice.

We find that both environmental stress and public environmental cues are associated with household green choices, but in opposing ways. First, hotter conditions are associated with lower green delivery adoption: a one-degree Celsius increase in abnormal temperature lowers take-up by about 0.36 to 0.43 percentage points, while a one-degree Celsius increase in wet-bulb temperature is associated with a 0.92 to 1.21 percentage point decline. Poorer air quality is also associated with lower green take-up, although the magnitude is smaller: a 10% increase in PSI lowers green take-up by about 0.12 percentage points. These results suggest that discomfort and cognitive strain may shift attention away from contributions to collective goods. Second, environment-related public events are associated with higher green delivery uptake. In our setting, event days increase

¹Clean & Green Singapore is an annual national campaign launched in 1990 to promote environmental awareness, tree planting, and community participation in waste reduction and energy conservation. It is typically anchored by large-scale public events, exhibitions, and school programs coordinated by the National Environment Agency (NEA). Go Green SG is a more recent initiative, launched in 2022 as a whole-of-nation sustainability movement under Singapore’s Green Plan 2030. It brings together a month-long series of outreach activities across government agencies, firms, and communities, emphasizing everyday sustainable actions. Both campaigns attract extensive media coverage and public engagement, temporarily increasing the visibility of environmental issues across households and firms.

green delivery uptake by 5 to 7 percentage points. This pattern is consistent with recent evidence that climate-related media coverage can shape everyday behavior (Beattie 2025) and that correcting misperceived pro-climate social norms can increase willingness to act against climate change (Andre et al. 2024). Third, environmental stress weakens the event-day association. A one-degree Celsius increase in wet-bulb temperature reduces the event-day association by about 3.4 percentage points, and a 10-point increase in PSI reduces it by about 3.5 percentage points. More broadly, these findings suggest that the effectiveness of public environmental cues may depend on consumers’ cognitive state at the moment of choice.

In addition, we show that the transaction environment itself shapes whether households act on environmental goals. Green take-up rises when more delivery slots are labeled as low-carbon, consistent with a broad choice-set exposure effect. In contrast, green adoption falls for higher-value and more complex orders, where households are likely to focus more on coordination and convenience, leaving less attention for environmental attributes. These findings reinforce our interpretation that sustainable behavior depends not only on preferences, but also on how attention is allocated within the decision environment. This interpretation is further supported by additional evidence. In the main results, we document parallel stress effects for both heat and air pollution, and in robustness checks using alternative temperature constructions and Google Trends-based proxies for environmental attention, we continue to find patterns consistent with an attention-based mechanism rather than a narrow feature of the baseline specification.

To formalize these findings, we rely on a simple behavioral inattention framework. A central challenge in sustainable consumption is that the benefits of pro-environmental actions are delayed and non-monetary, while the costs are immediate and salient (Gabaix 2019; DellaVigna 2009). Our model formalizes the idea that households allocate scarce attentional resources across competing choice dimensions, and that this allocation responds to both environmental conditions and public environmental cues. Physical stressors such as heat and air pollution may narrow attention and generate a tunneling effect (Mullainathan and Shafir 2013), crowding out consideration of environmental benefits—consistent with the decline in green delivery adoption we observe under adverse conditions. Conversely, environment-related public events may increase the perceptual prominence of environmental attributes (Bordalo et al. 2013), consistent with the immediate rise in adoption on event days. The framework further predicts that public environmental cues will be less effective when cognitive bandwidth is constrained (Mani et al. 2013), which is consistent with the weaker event-day association under high environmental stress. Together, these mechanisms provide a unified interpretation: household contributions to climate mitigation depend not only on preferences or incentives, but also on how attention is allocated at the moment of choice.

Our findings have broader implications for how household sustainability choices are shaped. First, they show that even low-cost green actions can be highly sensitive to attention at the moment of choice. In our setting, households are less likely to choose a labeled low-carbon delivery option when environmental stress is higher, even though doing so involves no direct monetary sacrifice. Second, the results suggest that the key constraint is not merely whether pro-environmental information is available, but whether it is used when decisions are made. This speaks to the distinction between frictions and mental gaps in information use emphasized by [Handel and Schwartzstein \(2018\)](#). Third, the findings show that public environmental cues can increase green behavior, but that their effectiveness depends on the cognitive conditions under which choices are made. The same public cue is less effective under greater heat stress.

Related Literature. This paper connects to three strands of research. First, it contributes to the literature on household sustainability choices and the role of information and choice architecture. Prior work shows that feedback, social comparisons, and nudges can meaningfully affect environmentally relevant routine decisions, including reduced cutlery use in food delivery ([He et al. 2023](#)), household energy use ([Jesso and Rapson 2014](#); [Allcott and Rogers 2014](#)), and responses to eco-labels, monetary incentives, and moral suasion ([Houde 2018](#); [Ito et al. 2018](#)). We extend this literature by using transaction-level data from an online grocery platform to study real-time adoption of a labeled ‘Less CO₂’ delivery option that entails a direct convenience trade-off but no monetary incentive. This setting allows us to examine how public environmental cues and adverse environmental conditions jointly shape green choice. It also allows us to test whether the event-day association is weaker when environmental stress constrains attention. In doing so, the paper also speaks to recent work on the drivers of household carbon emissions. While [Lyubich \(2025\)](#) shows that both individual characteristics and place-based factors shape variation in carbon footprints, we identify within-person shifts in a specific low-carbon choice associated with weather conditions, environment-related public events, and platform choice architecture.

Second, we relate to work on environmental conditions and economic behavior. Two environmental conditions are especially relevant for our setting: heat and air pollution. A large literature shows that heat affects cognition, high-stakes performance, judgment, and economic activity ([Dell et al. 2012](#); [Zivin et al. 2020](#); [Zivin and Neidell 2012](#); [Park et al. 2020](#); [Heyes and Saberian 2019](#)). A parallel literature shows that air pollution impairs cognitive performance and worker productivity, and that pollution information shapes protective behavior ([Zhang et al. 2018](#); [Chang et al. 2019](#); [Barwick et al. 2024](#); [Chang et al. 2018](#)). These findings make heat and air pollution natural measures of environmental stress in our context. We build on these two strands by studying how both stressors shape pro-environmental choices in a high-frequency setting. Our contribution is

to move beyond outcomes such as output, test scores, or judgment quality and show that the same environmental conditions can also weaken everyday contributions to collective climate mitigation. More broadly, our results suggest that climate-related stress may not only increase demand for private protection, but also weaken everyday forms of cooperative mitigation.

Third, we contribute to the literature on limited attention, cognitive scarcity, and decision-making. A central insight in this literature is that when cognitive resources are scarce, individuals overweight immediate and salient attributes and underweight delayed consequences (DellaVigna 2009; Gabaix 2019; Mullainathan and Shafir 2013; Shah et al. 2012). This framework has been used to explain systematic departures from full optimization across domains such as consumption, finance, and self-control. We extend it to a setting of everyday environmental choice and show that the attention channel helps explain not only whether households take green actions, but also when public environmental cues are more or less associated with those actions.

The paper proceeds as follows. Section 2 describes the background and data. Section 3 presents the empirical results. Section 4 interprets these results using a simple behavioral inattention framework. Section 5 presents robustness checks. Section 6 concludes.

2 Background and Data

This section describes our empirical setting and data. Section 2.1 introduces the institutional setting and administrative transaction data, including RedMart’s green delivery mechanism and the main features of the Singapore context. Section 2.2 describes the meteorological and environmental data used to construct district-level measures of heat exposure and region-level measures of air pollution. Section 2.3 presents our hand-collected database of environment-related public events. Together, these data enable us to link high-frequency variation in environmental stress and public environmental cues to revealed household behavior at the point of purchase.

2.1 Setting and Administrative Data

To study how attention shapes sustainable household choices, we combine a unique dataset of grocery purchases with variation in local weather conditions and environment-related public events. This setting allows us to observe green behavior at high frequency and to distinguish environmental stress from public environmental cues at the moment decisions are made.

Our administrative transaction data come from RedMart, an online grocery platform

in Singapore operated through Lazada under Alibaba Group. Similar to services such as Amazon Fresh, RedMart allows households to schedule deliveries of fresh produce, dairy, meat, and other household essentials through a mobile app. This setting is useful for our purposes because it provides detailed, high-frequency records of household choices in a context where customers choose among delivery slots that may differ in convenience and environmental attributes. It therefore allows us to study low-carbon delivery adoption using revealed-preference data from a real-world checkout environment.

The sustainable behavior we study is RedMart’s green delivery choice. After selecting a delivery address, customers are prompted at checkout to choose a delivery time slot, as illustrated in Figure 1. A slot is labeled ‘Less CO₂’ once at least one other customer in the same area has already selected that slot, creating scope for RedMart to consolidate deliveries and reduce emissions. Customers observe the environmental label, but not the underlying routing or consolidation logic. We define a green delivery decision as the choice of a delivery slot carrying this label. Figure 1 shows one example of the checkout interface. The set of labeled slots varies across transactions and is not tied to fixed delivery windows. Importantly, the Less CO₂ label does not provide a direct financial incentive. Delivery pricing consists of a fixed shipping fee and a slot fee that varies with urgency and flexibility. In our main analysis, we restrict attention to zero-fee slots, so the relevant trade-off is between delivery timing convenience and the environmental attribute highlighted by the label.

[Insert Figure 1]

Our sample begins with 1,000,000 delivery orders randomly drawn from RedMart between March 2023 and March 2025, with all records anonymized to protect customer privacy. The raw data include the delivery menu shown to each customer at checkout, the customer’s selected delivery slot, order characteristics, and coarse geographic information based on the delivery postcode. We apply two sample restrictions. First, to remove direct price confounds, we restrict attention to orders with zero slot fees, so that differences in green delivery choice are not mechanically driven by variation in delivery-slot prices. Second, we focus on orders in which customers were shown at least one delivery slot labeled Less CO₂. This leaves a working sample of roughly 325,000 orders in which customers faced an actual opportunity to choose green delivery.

Panel A of Table 1 reports summary statistics for the resulting sample. The dataset contains information along three main dimensions. First, it records the menu of delivery time slots shown to the customer at checkout. *Avail entries* denotes the total number of delivery slots displayed to the customer at the time of choice. *Eco avail entries* denotes the number of displayed slots that are labeled Less CO₂. Using these two variables, we construct our main menu-based measure, *Proportion*. It equals the number of Less CO₂

slots shown to the customer divided by the total number of displayed delivery slots. In the resulting sample, customers are shown an average of 2.250 green slots, and *Proportion* averages 25.1%, with substantial variation across orders (SD = 19.5%, range: 0.5%–100%). This variation is useful for identification because, conditional on our fixed effects and controls, it captures fluctuations in the displayed choice set that are less likely to be driven by time-invariant individual environmental preferences.

[Insert Table 1]

Second, the dataset contains detailed order characteristics, including expenditure, basket size, and weight. *Expensive* is a binary indicator equal to one if total order value is at least S\$100 and zero otherwise; 57.1% of orders in our sample are classified as high-value. We use this cutoff to distinguish higher-stakes transactions from routine smaller purchases. *Qty* measures basket size in units and is defined as the sum of item quantities in the order. *Order weight* records the total weight of the basket; in the regressions, we scale this variable in metric tons for numerical convenience. These variables capture different aspects of the private stakes and complexity of the checkout decision. Higher-value orders may increase concerns about delivery reliability and timing, while larger or heavier baskets may make coordination and convenience more salient. In these cases, the environmental attribute may receive less attention when immediate logistical concerns become more important. Including these variables also helps address compositional explanations by controlling for order characteristics that may correlate with green-slot availability or with changes in shopping behavior.

Third, the dataset records the customer’s realized delivery choice. *Green* is a binary indicator equal to one if the customer selects a green delivery slot. This is the main dependent variable in our analysis and provides a high-frequency revealed-preference measure of low-carbon delivery adoption. It exhibits substantial variation, with 40.8% of orders selecting the green option. The data also include *Delivery postcode 2d*, which records the first two digits of the delivery postcode and provides geographic information while preserving user anonymity.²

2.2 Meteorological Data

Next, we measure local environmental conditions using several datasets: the Historical Daily Records Dataset developed by the Meteorological Service Singapore, high-frequency

²Singapore postal codes contain six digits, with the first two digits identifying a postal sector. There are 81 such two-digit postal sectors in use, running from 01 to 82 with 74 omitted. These sectors map into 28 broader postal districts according to SingPost’s official List of Postal Districts. Using only the first two digits preserves meaningful spatial variation across locations while avoiding disclosure of precise household addresses. This level of aggregation is useful in our setting because it allows us to retain neighborhood-level variation while linking transactions to district-level environmental conditions.

weather data from Singapore’s open data portal, and daily air pollution data from the same portal. These data are especially useful in Singapore’s tropical setting, where day-to-day variation in environmental conditions can still generate economically meaningful behavioral responses (Fesselmeier et al. 2024).

The first weather dataset provides daily observations from 22 weather stations across Singapore beginning in 1980. It includes daily maximum and mean temperature, with most records available from 2011 onward. We use the 2011–2022 historical records to construct station-level seasonal norms for our abnormal temperature measure. The second dataset is a high-frequency weather dataset that records air temperature and relative humidity at the weather-station level at five-minute intervals beginning in 2016. These observations allow us to construct measures of current heat exposure, including mean temperature, maximum temperature, and wet-bulb temperature.

To link weather conditions to delivery locations, we aggregate station-level readings to the district level using a nearest-station procedure. Specifically, we map each of Singapore’s 28 postal districts to its four closest weather stations and average their readings. We then match each district to the corresponding two-digit postal sectors in the transaction data using SingPost’s official postal-district correspondence. This procedure allows us to merge district-level environmental conditions to each delivery order while preserving the spatial aggregation used to protect customer anonymity.

We also obtain the Historical Pollutant Standards Index dataset from Singapore’s open data portal for 2023–2025. The dataset reports daily PSI by region (east, west, north, south, and central). For our analysis, we match region-level PSI to the postal-sector level. Weather variables are assigned through the sector-to-district mapping, while PSI is assigned through the sector-to-region mapping.

Our analysis focuses on two key environmental measures. The first is heat. We adopt four complementary indicators of heat exposure, each capturing a distinct dimension of heat stress. These include mean temperature, maximum temperature, abnormal temperature, and wet-bulb temperature (Twb). Mean temperature and maximum temperature are daily mean and maximum temperatures (in °C) on the purchase day. Mean temperature captures sustained heat exposure throughout the day, while maximum temperature isolates peak heat stress during the hottest hours.

We construct abnormal temperature as the deviation of the observed daily maximum temperature from a station-specific climatological norm. Using historical data from 2011–2022, we compute the norm for each station and calendar date as the average daily maximum temperature within a ± 5 -day window around that calendar date, combining observations across all years in 2011–2022. We then define abnormal temperature on the purchase day as the station’s observed daily maximum minus this calendar-date norm, and

we aggregate to the district level by averaging the values from the two nearest weather stations. In Section 5, we verify robustness using an alternative measure that instead averages the four nearest stations, providing broader spatial smoothing. This measure isolates unexpected heat shocks from predictable seasonal patterns, capturing heat events to which households are less physiologically and behaviorally adapted.

Wet-bulb temperature (T_{wb}) captures physiological heat stress arising from the combination of heat and humidity. To calculate this temperature, we first obtain 5-minute observations of temperature (T , in $^{\circ}\text{C}$) and relative humidity (RH , in $\%$) for each weather station. Then, to match the postal level data in our main dataset, we use the nearest four stations and get district-level data by averaging the four values of temperature and humidity. We use the following approximation (Stull 2011).³ This measure is particularly informative in Singapore’s tropical climate, where high humidity frequently amplifies the physiological burden of heat beyond what dry-bulb temperature alone would suggest.

The second is air pollution, measured using Singapore’s Pollutant Standards Index (PSI), the country’s official air-quality indicator published daily by the National Environment Agency. Similar index-based measures (e.g., API/AQI) are widely used in economics to study pollution information, avoidance behavior, and household responses (Chang et al. 2018; Barwick et al. 2024). PSI aggregates six major pollutants, including $\text{PM}_{2.5}$, PM_{10} , SO_2 , NO_2 , O_3 , and CO , into a single publicly communicated risk metric. We use PSI for three reasons. First, it is the policy-relevant measure that households and media directly observe in real time, which aligns with our attention-based mechanism. Second, as a composite indicator, it captures overall ambient pollution burden rather than variation in only one pollutant. Third, daily region-level PSI provides high-frequency spatial variation that can be consistently matched to each order using delivery location and date. Values below 50 are classified as ‘good’, 51 to 100 as ‘moderate’, 101 to 200 as ‘unhealthy’, and higher values as ‘very unhealthy’ or ‘hazardous’. To link PSI to household choices, we map each two-digit postal sector in the transaction data to one of Singapore’s five PSI reporting regions and assign the corresponding region-day PSI value. Since Singapore does not have official, standardized geographic boundaries for these five regions, we map postal sectors to regions using NEA’s regional PSI reporting convention, Urban Redevelopment Authority district definitions, and common geographic boundaries. In our dataset, all postal sectors within a region receive the same PSI value on each calendar day from 2023 to 2025.

Table 1 Panel B summarizes the meteorological, air-quality, and event variables. District-day wet-bulb temperature (T_{wb}) averages 25.069 degrees Celsius with a standard

³Wet-bulb temperature (T_w) approximation in Stull (2011):

$$T_w \approx T \cdot \arctan(0.151977\sqrt{RH} + 8.313659) + \arctan(T + RH) - \arctan(RH - 1.676331) + 0.00391838 RH^{3/2} \cdot \arctan(0.023101 RH) - 4.686035.$$

deviation of 0.751, ranging from 22.01 degrees Celsius to 27.54 degrees Celsius. Appendix Section B documents meaningful spatial and temporal variation in district-level wet-bulb temperature across Singapore. Air quality (*PSI*), measured by region-day PSI, averages 45.424 with a standard deviation of 11.972, and ranges from 14.292 to 108.375. The same-day environment-related public-event indicator (*HasEvent*) has a mean of 0.086 and a standard deviation of 0.280, with values from 0 to 1.

2.3 Environment-Related Public Events Data

In addition to weather shocks, we compile a dataset of environment-related public events in Singapore between 2023 and 2025. We systematically hand-collected these events from official announcements by the National Environment Agency (NEA), the Ministry of Sustainability and the Environment (MSE), and affiliated public programs. The resulting list includes sustainability campaigns and announcements (e.g., *Clean & Green Singapore Day*, *Go Green SG*, *World Cleanup Day*) as well as salient environment- or weather-related disruptions and public updates, such as dry spells, monsoon advisories, and oil-spill cleanup updates. Each event is coded with its start and end dates, scope, and category.

To integrate these events into our empirical analysis, we expand each event into a daily panel, matching the event period to the transaction-level dataset by calendar date. We then construct an indicator variable, *HasEvent*, equal to one on days when at least one environment-related public event took place, and zero otherwise. This construction captures periods when environmental issues were more visible in public information, but it should not be interpreted as a measure of public campaigns alone. Table A1 in the Online Appendix presents a listing of the events.

Our event dataset allows us to connect publicly visible environmental information to high-frequency consumer choice data. Prior work has often relied on survey-based measures of environmental attitudes or cross-sectional exposure to policy initiatives. By contrast, our hand-collected day-level record allows us to study short-run variation in exposure to environment-related public events. Because many of these events are nationwide and publicly documented, their timing is less likely to be driven by individual transaction-level choices. Their short duration and clear documentation provide useful within-period variation for testing whether consumer choices are associated with public environmental cues.

3 Empirical Results

We now present the main empirical results. We begin by examining whether environmental stress reduces green delivery adoption, focusing first on heat exposure in Section 3.1 and then on air pollution in Section 3.2. Section 3.3 turns to public environmental cues by studying whether environment-related public events are associated with higher adoption and whether their effects depend on contemporaneous environmental stress. Finally, Section 3.4 examines how features of the transaction environment, including menu composition and order characteristics, shape green delivery choice.

3.1 Heat and Green Delivery

We begin by testing the impact of heat exposure to validate whether environmental stressors systematically reduce sustainable behavior. This section examines four complementary measures of heat exposure that theory suggests should crowd out environmental considerations in household decision-making.

Our analysis employs abnormal temperature (deviations from seasonal norms), mean temperature, maximum temperature, and wet-bulb temperature. Each captures distinct dimensions of heat exposure that can impair cognitive performance and redirect attention toward immediate comfort. Abnormal temperature isolates the effect of unexpected temperature anomalies from seasonal patterns, to which households may be less adapted. Maximum temperature captures peak daily heat exposure when heat exposure is most acute, while mean temperature reflects sustained heat conditions throughout the day. Wet-bulb temperature proves particularly informative as it integrates heat and humidity to measure the body’s actual cooling capacity through perspiration, providing a direct indicator of physiological strain beyond what air temperature alone reveals. These measures allow us to test whether different manifestations of heat stress systematically reduce the weight households place on environmental considerations in daily consumption choices, with results consistently demonstrating that heat discomfort undermines sustainable behavior across all four metrics. We estimate our baseline specification as an order-level linear probability model with rich fixed effects.

$$\begin{aligned} \text{Green}_{oidt} = & \alpha + \beta_1 \text{Temp}_{dt} + \beta_2 \text{Proportion}_{oidt} + \beta_3 \text{OrderChar}_{oidt} \\ & + \gamma_i + \delta_d + \delta_{ym} + \delta_{dow} + \delta_h + \varepsilon_{oidt}. \end{aligned} \tag{1}$$

In this specification, Green_{oidt} equals one if order o placed by individual i in district d on date t selects a delivery slot labeled ‘Less CO₂’ at checkout (i.e., a consolidated low-emissions slot), and zero otherwise. The main explanatory variable Temp_{dt} captures alternative measures of heat exposure, including abnormal temperature, mean

temperature, maximum temperature, and district-day wet-bulb temperature. A negative β_1 implies that higher heat stress reduces the probability of choosing green delivery. Proportion_{oidt} measures green slot availability, and OrderChar_{oidt} includes order characteristics. The specification controls for OrderChar_{oidt} , which includes Expensive_{oidt} , Qty_{oidt} , and $\text{Order weight}_{oidt}$.

Table 2 presents our main findings on how heat stress undermines sustainable behavior. Because identification relies on within-buyer variation over time, buyers observed only once are dropped from the estimation sample, along with other singleton observations implied by the fixed-effects structure. We examine four complementary temperature measures, each capturing distinct dimensions of heat exposure that theory suggests should narrow attention toward immediate comfort. All four measures show negative and highly significant effects on green delivery choice. This consistency suggests that the result is not driven by a particular way of measuring temperature. Rather, the negative relationship appears to be a general feature of hotter conditions, regardless of whether heat is captured as an unexpected shock, average daily exposure, peak heat, or humidity-adjusted heat stress.

[Insert Table 2]

Abnormal temperature is especially informative because it is measured relative to historical seasonal norms, capturing unexpected heat shocks to which households may be less adapted. Column 1 shows that each additional degree Celsius of abnormal heat reduces green delivery adoption by 0.43 percentage points, with the estimate statistically significant at the 1% level. The effect remains similar after adding delivery characteristics in Column 2, where the coefficient is -0.0036 and remains statistically significant at the 1% level. This pattern is consistent with the interpretation that unusual heat shifts attention toward immediate comfort and convenience, reducing the weight placed on environmental considerations at checkout.

The same qualitative pattern holds when heat is measured using daily mean temperature, daily maximum temperature, and wet-bulb temperature in levels. These coefficients can be interpreted as the percentage-point change in green delivery adoption associated with a one-degree Celsius increase in temperature. For mean temperature, which captures sustained daily heat exposure, Column 2 implies that a one-degree Celsius increase is associated with a 0.32 percentage point decline in green delivery choice. For maximum temperature, which captures peak daily heat, the corresponding estimate is 0.24 percentage points. These estimates are statistically significant at the 1% level and point in the same direction as the abnormal-temperature result.

Among the four heat measures, wet-bulb temperature has the largest estimated association. This is consistent with wet-bulb temperature being a closer proxy for experienced

heat discomfort in Singapore, since it combines air temperature and humidity and therefore captures the physiological burden of heat more directly than dry-bulb temperature alone. A 1 °C increase in wet-bulb temperature is associated with a 0.92 percentage point decline in green delivery adoption after adding delivery characteristics, with the estimate statistically significant at the 1% level. Given that wet-bulb temperature in our sample ranges from roughly 22 °C to 27.5 °C, moving from relatively comfortable conditions of 23 °C to more stressful conditions of 27 °C implies a decline in green choice of about 3.7 percentage points. We interpret this pattern as consistent with heat-humidity stress shifting attention toward immediate comfort and convenience.

The negative estimates are consistent across all four temperature measures, whether capturing abnormal shocks, sustained exposure, peak heat, or humidity-adjusted heat stress. This pattern suggests that the heat result is not specific to one temperature construction. The larger estimate for wet-bulb temperature is consistent with the interpretation that combined heat-humidity stress is more closely related to experienced discomfort than dry-bulb temperature alone.

3.2 Air Pollution and Green Delivery

We next examine whether the same behavioral pattern extends to another environmental stressor: air pollution, using Singapore’s Pollutant Standards Index (PSI). As discussed in Section 2.2, PSI is Singapore’s official AQI-style indicator and is the pollution measure most directly observed by households. Table 3 examines whether air pollution affects green delivery choice through similar attention mechanisms as heat stress. Like heat, poor air quality generates physiological discomfort and has been documented to impair cognitive function, suggesting it may similarly crowd out environmental considerations. The specification parallels Equation (1), replacing the temperature variable with the natural logarithm of PSI.

$$\begin{aligned} \text{Green}_{oidt} = & \alpha + \beta_1 \log(\text{PSI}_{dt}) + \beta_2 \text{Proportion}_{oidt} + \beta_3 \text{OrderChar}_{oidt} \\ & + \gamma_i + \delta_d + \delta_{ym} + \delta_{dow} + \delta_h + \varepsilon_{oidt}. \end{aligned} \quad (2)$$

Here, the dependent variable Green_{oidt} equals one if order o placed by individual i in district d on date t is delivered via the green option. The key regressor $\log(\text{PSI}_{dt})$ denotes the natural logarithm of the Pollutant Standards Index in delivery district d on date t . Proportion_{oidt} is the share of green slots in the delivery menu. The specification controls for OrderChar_{oidt} , which includes Expensive_{oidt} , Qty_{oidt} , and $\text{Order weight}_{oidt}$ (total order weight).

To control for unobserved heterogeneity, the model includes individual fixed effects (γ_i), district fixed effects (δ_d), year-month fixed effects (δ_{ym}), day-of-week fixed effects (δ_{dow}), and slot-time fixed effects (δ_h). Standard errors are clustered at the individual level.

[Insert Table 3]

The coefficient is negative in both columns. Column 1, which excludes delivery controls, yields an estimate of -0.0117 , statistically significant at the 5% level. Column 2 adds controls for green slot availability and order characteristics, and the estimate is statistically significant at the 1% level. A 10% increase in PSI reduces green delivery choice by approximately 0.12 percentage points (Column 2). The stability of the coefficient across specifications suggests that the pollution effect does not operate through shifts in order composition or menu structure. To gauge economic magnitude, Singapore’s PSI typically ranges from around 25 on clean days to above 100 during regional haze episodes driven by Indonesian forest fires. Moving from clean conditions ($\text{PSI} \approx 25$) to moderately polluted conditions ($\text{PSI} \approx 80$), which more than triples the PSI level, implies a reduction in green choice of approximately 1.4 percentage points based on the Column 2 estimate. Relative to the baseline green-adoption rate of 40.8%, this corresponds to a decline of about 3.5%. The negative pollution effect is consistent with the view that poor air quality acts as an environmental stressor. Prior work shows that air pollution can impair cognitive performance and reduce productivity (Zhang et al. 2018; Chang et al. 2019), making it plausible that worse air quality shifts attention away from environmental attributes at the moment of choice.

3.3 Environment-Related Public Events and Green Delivery

Having shown that environmental stressors such as heat and air pollution are associated with lower green take-up, we next examine whether environment-related public events are associated with higher green delivery adoption. The attention-based interpretation suggests that when environmental issues become more visible in public discourse, households may place greater weight on the environmental attribute of delivery slots at checkout. We therefore study prominent environment-related public events in Singapore, including Earth Day, World Environment Day, Clean & Green Singapore, Go Green SG, climate-related public campaigns, major sustainability announcements, and salient environmental or weather-related public updates.

Singapore provides a useful setting for this analysis because environment-related public events are frequently documented by public agencies such as the National Environment

Agency and the Ministry of Sustainability and the Environment, and often receive coverage through traditional media, social media, and public outreach activities. These events may raise the visibility of environmental issues for a broad set of households, while not directly changing the price, technology, or checkout attributes of green delivery.

We interpret these events as public environmental cues rather than as a fully exogenous experiment. If attention to environmental issues matters for green delivery choice, green take-up should rise during these periods, with limited evidence of similar increases before the events and a decline once the public cue fades. The event-study analysis below examines these patterns.

3.3.1 Event Study

We estimate an event study to trace how green delivery adoption evolves around environment-related public events. Because these events can last multiple consecutive days, we group consecutive event days into a single event episode. We define the event period as the event days within an episode and construct relative-time indicators for the $K = 14$ days before an episode begins and the $K = 14$ days after an episode ends.

To avoid contamination from nearby episodes, we drop calendar days that fall into overlapping windows, namely days that are simultaneously within the pre-window of one episode and the post-window of another.⁴

For order o placed by individual i in district d on date t , we estimate

$$Green_{oidt} = \alpha + \sum_{\tau \in \mathcal{T}} \beta_{\tau} \mathbf{1}[\tau(t) = \tau] + \mathbf{X}'_{oidt} \pi + \gamma_i + \delta_d + \delta_{ym} + \delta_{dow} + \delta_h + \varepsilon_{oidt}, \quad (3)$$

where $Green_{oidt}$ equals one if the customer selects a green delivery slot. $\tau(t)$ denotes days relative to the episode, with $\tau = 0$ on event days, $\tau = -k$ for the k th day before the episode starts, and $\tau = +k$ for the k th day after the episode ends. We omit $\tau = -1$ as the reference period. $\mathbf{X}_{oidt}$ includes the baseline order and delivery-menu controls, and we absorb individual, district, year-by-month, day-of-week, and slot-hour fixed effects. Standard errors are clustered by buyer and date.

Figure 2 plots the estimated results and 90% confidence intervals for $t \in [-5, 5]$. The pattern reveals three key findings that support our attention-based framework.

[Insert Figure 2]

First, the event period ($t = 0$) is associated with an increase in green delivery choice. The event study coefficient at $t=0$ is approximately 4 percentage points, somewhat smaller

⁴Appendix C provides the formal construction of episodes, relative-time indicators, and the overlap criterion.

than the pooled estimate of 5 to 7 percentage points in Table 4. This difference likely reflects the event study’s narrower identification window. Both estimates indicate that environment-related public events are associated with higher green adoption relative to the unconditional baseline of 40.8%. This effect is statistically significant at the 10% level.

Second, coefficients in the pre-event window ($t = -5$ to $t = -2$) are small in magnitude and statistically indistinguishable from zero. This pattern is consistent with limited anticipation effects or differential pre-trends. Consumers do not appear to systematically alter their behavior in the days leading up to environment-related public events.

Third, post-event coefficients ($t = 1$ to $t = 5$) quickly return to near zero and lack statistical significance, suggesting that the behavioral response is temporary rather than a persistent shift in underlying preferences or values. This pattern is consistent with an attention-based interpretation: once media coverage subsides and the event recedes from public discourse, cognitive resources may return to their typical allocation across competing concerns.

3.3.2 Pooled Event Effects

To complement the event study and provide a summary measure of event associations, Table 4 estimates pooled regressions where *HasEvent* equals one for all days within event episodes. Event days are associated with a 7.03 percentage point increase in green delivery choice without controls and a 5.18 percentage point increase with controls, both significant at the 1% level. The magnitude in Column 2 corresponds to approximately 13% relative to baseline.

[Insert Table 4]

The event study provides evidence consistent with an attention-based interpretation. Periods when environmental issues become more visible are associated with a temporary rise in green delivery adoption, but the association dissipates quickly once the public cue fades. This pattern motivates the interaction analysis below, which asks whether environmental stress weakens the response to such public environmental cues.

3.3.3 Interaction of Environmental Stress and Public Events

The preceding results establish two patterns. Environmental stressors, both heat and air pollution, are associated with lower green delivery adoption. Environment-related public events, by contrast, are associated with higher adoption. A natural next question is whether these forces operate independently, or whether environmental stress weakens the

event-day association. Table 5 addresses this question by examining interactions between public events and environmental stressors.

[Insert Table 5]

We focus on the specifications in Table 5, which report the interactions between public events and wet-bulb temperature and air pollution, both measured in levels, with delivery controls. In both columns, the interaction between the event indicator and environmental stress is negative and precisely estimated, indicating that the positive event-day association is weaker under more adverse environmental conditions.

The largest attenuation appears for wet-bulb temperature in Column 1 of Table 5. A one-degree Celsius increase in wet-bulb temperature reduces the event-day association by about 3.41 percentage points, statistically significant at the 1% level. The main effect of wet-bulb temperature is also negative and statistically significant (-0.0044 , at the 10% level), indicating that humidity-adjusted heat is associated with lower green choice even outside event periods. At more comfortable conditions (wet-bulb temperature around 23°C), the implied event-day association is larger. But as wet-bulb temperature rises toward 27°C , the interaction offsets much of this association. This pattern is consistent with the idea that physiological heat stress, as captured by the body's reduced capacity for evaporative cooling, constrains the cognitive resources available for processing environmental information.

Column 2 of Table 5 extends the analysis to air pollution. The interaction between the event indicator and PSI is negative and statistically significant at the 1% level. A one-point increase in PSI reduces the event-day association by about 0.35 percentage points; equivalently, a 10-point increase reduces it by about 3.5 percentage points. The main effect of PSI is positive and statistically significant in this specification (0.0003), but this coefficient captures the conditional association between pollution and green choice on non-event days. One interpretation is that higher PSI can make environmental conditions more visible to consumers on ordinary days, while poor air quality may also generate discomfort or cognitive stress. When a public environmental cue is already present, the negative interaction indicates that higher pollution weakens the event-day association with green delivery adoption. On event days, the net PSI association is negative because the interaction term more than offsets the main PSI coefficient.

The supplementary appendix tables show that this attenuation pattern is not specific to these two preferred specifications. Table A2 shows that the qualitative pattern is similar in the corresponding level specifications without delivery controls. Table A3 reports the corresponding controlled results for abnormal temperature, mean temperature, and maximum temperature in levels. Across all three measures, the interaction

between the event indicator and temperature is negative and statistically significant at the 1% level. For abnormal temperature in Column 1 of Table A3, each additional degree Celsius of unexpected heat reduces the event-day association by about 2.96 percentage points. Given that the unconditional event-day coefficient is approximately 5-7 percentage points (Table 4), an abnormal temperature shock of two degrees Celsius would offset much of the event-day association. For mean temperature in Column 2 of Table A3, a one-degree Celsius increase reduces the event-day association by about 1.37 percentage points. For maximum temperature in Column 3 of Table A3, the corresponding figure is 1.10 percentage points. Table A4 reports the corresponding log specifications with controls.

Two features of these results are worth noting. First, the attenuation pattern holds across qualitatively different stressors. Temperature and air pollution impose distinct forms of physiological burden, yet both weaken the event-day association. This generality is consistent with a common channel, plausibly through reduced cognitive bandwidth, rather than a pattern tied to any single stressor. Second, the interaction effects are meaningful relative to the unconditional event-day coefficient of 5-7 percentage points. Moderate increases in environmental stress can offset a share of the association between public environmental cues and green delivery adoption. Section 4 develops a framework that formalizes this interaction between stress and public environmental cues.

3.4 Internal Drivers of Green Delivery Choice

Having examined how external stressors and public environmental cues are associated with green behavior, we next consider whether the transaction environment itself also affects whether households choose green delivery. We focus on menu composition and order characteristics as internal features of the decision environment that may shape visibility, scheduling costs, and immediate convenience concerns.

We estimate the following transaction-level regression to identify the key internal determinants:

$$\text{Green}_{oidt} = \alpha + \beta_1 \text{Proportion}_{oidt} + \beta_2 Z_{oidt} + \gamma_i + \delta_d + \delta_{ym} + \delta_{dow} + \delta_h + \varepsilon_{oidt}, \quad (4)$$

where Green_{oidt} equals one if order o chooses a delivery slot labeled ‘Less CO₂’ at checkout and zero otherwise. The key variable Proportion_{oidt} measures the share of green slots available in the choice menu. We interpret this variable as a broad choice-set exposure measure rather than as salience alone. A higher green-slot share may make the Less CO₂ option more visible, lower search or scheduling costs, and mechanically expand the set of feasible green choices. Z_{oidt} represents order characteristics, which in different columns of

Table 6 is defined alternatively as Expensive_{oidt} , Qty_{oidt} , or $\text{Order weight}_{oidt}$. The model includes individual fixed effects (γ_i) to control for time-invariant consumer heterogeneity in environmental preferences, along with district, year-month, day-of-week, and time-slot fixed effects. Standard errors are clustered at the individual level.

Table 6 presents results examining how menu design and order characteristics shape green delivery adoption. Each column isolates a different internal driver while maintaining the comprehensive fixed effects structure to ensure identification comes from within-individual variation over time.

[Insert Table 6]

Column 1 illustrates the role of green-slot share in the delivery menu. A 10 percentage point increase in the share of green slots available is associated with a 4.8 percentage point increase in the probability of choosing green delivery, statistically significant at the 1% level. Given that the baseline green adoption rate is around 40.8%, this represents roughly a 12% increase relative to baseline. We interpret this coefficient as a broad choice-set exposure effect. When green options make up a larger share of the choice set, they may become more visible, easier to locate, and more likely to include a convenient delivery time. The coefficient should therefore not be read as isolating salience alone; it may also reflect lower scheduling costs and mechanical expansion of feasible green slots.

Column 2 examines whether customers choose green delivery differently across order values. The coefficient on whether the order value is high is statistically significant at the 1% level, indicating that when the same individual places a high-value order (total price $\geq$ S\$100), they are 7.42 percentage points less likely to choose green delivery compared to their lower-value orders. One interpretation is that higher-value orders make delivery reliability and timing more important, reducing willingness to accept any perceived scheduling constraint associated with the green option. The negative coefficient on order value is consistent with this convenience-cost interpretation, although it need not imply a change in underlying environmental preferences.

Column 3 examines how order complexity is associated with green choice. Each additional item in the basket reduces green delivery probability by 0.29 percentage points, statistically significant at the 1% level. For a one-standard-deviation increase in quantity, this translates to a 4.4 to 4.6 percentage point reduction in green choice, an 11% decrease relative to baseline adoption rates. This pattern is consistent with larger baskets increasing logistical concerns and leaving less room for secondary attributes such as environmental labeling at checkout.

Column 4 adds order weight as a predictor of green delivery adoption. A one-standard-deviation increase in order weight (approximately 24 kg) reduces the probability of choos-

ing green delivery by approximately 2.1 percentage points, statistically significant at the 1% level. Households with heavier shipments may also be more concerned about delivery timing to ensure they are home to receive bulky items, making the flexible scheduling required by green delivery less attractive.

4 Behavioral Framework

To interpret the observed patterns, we propose a behavioral inattention framework of green delivery choice. The central idea is that the environmental benefit of green delivery is delayed and non-monetary, whereas the convenience cost is immediate and directly experienced. The gap between stated environmental concern and realized green behavior may therefore depend not only on preferences, but also on how much attention households allocate to future environmental consequences at the moment of choice.

Our framework integrates two forces that operate on this attentional margin. Physical stress can both heighten awareness of environmental deterioration and narrow cognitive bandwidth toward immediate comfort, convenience, and effort costs. Environment-related public events provide a separate public-information cue that may make environmental considerations more prominent at the moment of choice. The reduced-form model below allows these channels to coexist. It is intended as an organizing framework for the empirical patterns rather than as a structural estimation exercise.

4.1 Setup

4.1.1 The Choice Environment

Consider a household making a discrete choice $a \in \{G, N\}$ between Green delivery (G) and Non-green delivery (N). In our setting, both options are restricted to zero-fee slots, so the relevant trade-off is between scheduling convenience and environmental consequences.

The menu presented to the household contains n delivery slots in total, of which a fraction $\phi \in [0, 1]$ are green slots. We interpret ϕ as green-slot share, corresponding to the *Proportion* variable in the data. In the sample, *Proportion* has mean 25.1%, standard deviation 19.5%, and range 0.5% to 100%. This wide range reflects variation in menu composition and implies that the green share of the menu may shape the ease with which households identify and select a green slot.

4.1.2 Utility Specification

A fully attentive household derives utility from two dimensions of option a : immediate scheduling convenience and future environmental benefit. The rational utility of option a is

$$U(a) = u_0(a) + \delta u_1(a), \quad (5)$$

where $u_0(a)$ is immediate utility, $u_1(a)$ is the utility from the environmental consequence, and $\delta \in (0, 1)$ is the discount factor. We impose two assumptions: $u_0(G) < u_0(N) \leq 0$, so green delivery carries an immediate scheduling cost, and $u_1(G) > u_1(N) \geq 0$, so it yields a larger environmental benefit through route consolidation.

4.1.3 Interpretation of the Environmental Term

The term $u_1(a)$ captures delayed environmental utility rather than immediate expressive rewards. Let

$$\Delta u_1 \equiv u_1(G) - u_1(N) > 0$$

denote the household's long-run valuation of the emissions reduction embedded in the green option. Because this benefit is delayed and diffuse, it is less cognitively accessible at checkout than the immediate scheduling cost.

4.1.4 Attention-Weighted Utility and Choice-Set Exposure

Following [Gabaix \(2014\)](#), boundedly rational households place weight $m \in [0, 1]$ on the future environmental dimension, yielding

$$U^s(a) = u_0(a; \phi) + m \delta u_1(a). \quad (6)$$

We allow green-slot share to shift the immediate component of the green option:

$$u_0(G; \phi) = u_0(G) + \gamma \phi, \quad u_0(N; \phi) = u_0(N), \quad \gamma > 0. \quad (7)$$

A larger green share can make the green option easier to identify, lower search or scheduling costs, and mechanically expand the set of feasible green choices. Thus, ϕ captures broad choice-set exposure rather than salience alone.

Let

$$\Delta u_0(\phi) \equiv u_0(G; \phi) - u_0(N) = [u_0(G) - u_0(N)] + \gamma \phi. \quad (8)$$

The household chooses green delivery if and only if

$$\Delta u_0(\phi) + m \delta \Delta u_1 > 0. \quad (9)$$

With a Type-I Extreme Value shock, the resulting choice probability is

$$P(G) = \Phi(m, \phi) = \frac{1}{1 + \exp(c_0 - \gamma\phi - m \delta \Delta u_1)}, \quad (10)$$

where $c_0 \equiv u_0(N) - u_0(G) > 0$ is the baseline scheduling cost of green delivery. The comparative statics below are stated in terms of $P(G)$ and therefore map directly to the estimated percentage-point effects.

4.2 Attention Rule and Empirical Mapping

This subsection maps the model objects to the empirical variation used above. The mapping is reduced-form and should not be read as a structural recovery of attention. The stress variable S corresponds to the environmental stress proxy in a given specification, such as wet-bulb temperature or PSI. The event indicator E corresponds to *HasEvent*, which marks days with at least one environment-related public event. The choice-set exposure variable ϕ corresponds to *Proportion*, the share of displayed delivery slots labeled Less CO₂.

4.2.1 Functional Form of Attention

The attention weight m on the environmental dimension is determined by centered environmental stress $\tilde{S} \equiv S - \bar{S}$ and public events $E \in \{0, 1\}$. Empirically, we use district-day stress proxies in separate specifications, with S measured using wet-bulb temperature or PSI in the corresponding functional form of the empirical table. Centering makes $\bar{m}$ interpretable as baseline attention at average stress.

We adopt a parsimonious reduced-form specification:

$$m(\tilde{S}, E) = \bar{m} + \sigma E - \alpha \tilde{S} - \beta(\tilde{S} \times E), \quad (11)$$

where $\bar{m} \in (0, 1)$ is baseline attention, $\sigma > 0$ is the public-event attention gain at average stress, $\alpha > 0$ is the net stress effect on attention outside event days, and $\beta > 0$ captures how stress attenuates the event-related attention gain. We study Equation (11) over the observed support of $\tilde{S}$ and restrict parameters so that $m(\tilde{S}, E) \in (0, 1)$ throughout that support.

Equation (11) is a reduced-form maintained assumption rather than a full information-

allocation solution. It is motivated by bounded-attention and rational-inattention frameworks (Gabaix 2014; Sims 2003), together with evidence that scarcity-like stress can compress effective cognitive bandwidth (Mani et al. 2013; Shah et al. 2012; Mullainathan and Shafir 2013).

4.2.2 Qualitative Properties of the Attention Function

Property 1 (Net stress effect). For both event and non-event days,

$$\frac{\partial m}{\partial \tilde{S}} = -\alpha - \beta E < 0. \quad (12)$$

Property 2 (Event-related gain at fixed stress). At any given stress level,

$$m(\tilde{S}, 1) - m(\tilde{S}, 0) = \sigma - \beta \tilde{S}. \quad (13)$$

Event-Attention Condition E. Over the observed support of $\tilde{S}$,

$$\sigma > \beta \tilde{S}^{\max}, \quad (14)$$

where $\tilde{S}^{\max}$ is the sample maximum of centered stress. Under Condition E, Equation (13) is strictly positive throughout the sample support.

Property 3 (Stress attenuates event-related gain).

$$\frac{\partial}{\partial \tilde{S}} [m(\tilde{S}, 1) - m(\tilde{S}, 0)] = -\beta < 0. \quad (15)$$

4.2.3 Competing Mechanism and Parsimonious Net-Effect Specification

Conceptually, environmental stress may activate both a positive public-awareness channel and a negative bandwidth channel, through which households focus more on immediate comfort and convenience. In the main text, we do not model these channels separately. Instead, Equation (11) captures their net effect in reduced form through (α, β) . This is consistent with evidence that thermal and pollution-related stress can impair higher-order cognitive processing in ways that are particularly relevant for deliberative trade-off decisions (Shibasaki et al. 2017; Graff Zivin et al. 2018).

4.3 Equilibrium and Comparative Statics

For convenience, let

$$\Phi_m \equiv \frac{\partial \Phi}{\partial m} = \delta \Delta u_1 \Phi(1 - \Phi) > 0, \quad (16)$$

$$\Phi_{mm} \equiv \frac{\partial^2 \Phi}{\partial m^2} = (\delta \Delta u_1)^2 \Phi(1 - \Phi)(1 - 2\Phi), \quad (17)$$

$$\Phi_\phi \equiv \frac{\partial \Phi}{\partial \phi} = \gamma \Phi(1 - \Phi) > 0. \quad (18)$$

Note that $\Phi_{mm} > 0$ if and only if $\Phi < 1/2$.

Proofs. The proofs of Propositions 1–4 are presented in Appendix D.

Proposition 1 (Environmental Stress Reduces Green Choice). *Let $m = m(\tilde{S}, E)$ be given by Equation (11). Then*

$$\frac{\partial P(G)}{\partial \tilde{S}} < 0.$$

Proposition 2 (Public Events Increase Green Choice). *Under Event-Attention Condition E, for any fixed $\tilde{S}$ on the observed support,*

$$P(G \mid E = 1, \tilde{S}) - P(G \mid E = 0, \tilde{S}) > 0.$$

Proposition 3 (Environmental Stress Attenuates the Event Effect). *Under Event-Attention Condition E, at any $\tilde{S}$ in the observed support such that*

$$P(G \mid E = 0, \tilde{S}) < \frac{1}{2} \quad \text{and} \quad P(G \mid E = 1, \tilde{S}) < \frac{1}{2},$$

the event premium

$$\Delta_E P(G; \tilde{S}) \equiv P(G \mid E = 1, \tilde{S}) - P(G \mid E = 0, \tilde{S})$$

is decreasing in $\tilde{S}$.

Proposition 4 (Choice-Set Exposure Increases Green Choice). *For any $\gamma > 0$,*

$$\frac{\partial P(G)}{\partial \phi} = \gamma \Phi(1 - \Phi) > 0.$$

Thus, a larger green share in the delivery menu increases the probability of green choice.

4.4 Discussion and Empirical Interpretation

The propositions organize the empirical patterns documented above. Higher heat and pollution are associated with lower green take-up; event days are associated with higher

take-up; and the event-day association is weaker when environmental stress is higher. In the main heat results, a one-degree Celsius increase in wet-bulb temperature is associated with a 0.92 to 1.21 percentage point decline in green delivery adoption. In the pollution results, a 10% increase in PSI is associated with about a 0.12 percentage point decline. These patterns are consistent with Proposition 1.

Event days are associated with a 5 to 7 percentage point increase in green delivery adoption, consistent with Proposition 2. The interaction estimates are consistent with Proposition 3: the event-day association is weaker when heat-humidity stress or air pollution is higher. We interpret this pattern as consistent with environmental stress reducing the effectiveness of public environmental cues, rather than as definitive evidence on a single psychological channel.

Finally, the positive coefficient on *Proportion* is consistent with Proposition 4, but it should be interpreted carefully. A higher green-slot share may increase visibility, lower search or scheduling costs, and mechanically expand the set of feasible green choices. Thus, *Proportion* captures broad choice-set exposure rather than salience alone. Relative to [Gabaix \(2014\)](#) and [Sims \(2003\)](#), our contribution is not a new micro-foundation for attention, but an empirically disciplined reduced-form framework linked to high-frequency household choice.

The framework also aligns with the within-buyer identification strategy in the empirical analysis. Household fixed effects absorb time-invariant heterogeneity in environmental valuation, discounting, and scheduling costs, while within-household variation in district-day stress and event timing shifts $m(\tilde{S}, E)$ and therefore shifts $P(G)$. Under this interpretation, the empirical coefficients capture within-household associations between stress, public environmental cues, choice-set exposure, and green delivery adoption at the moment of choice.

5 Robustness

This section evaluates the robustness and broader validity of our findings. We proceed in two parts. Section 5.1 presents complementary evidence from Google Trends, which helps validate the interpretation of environment-related public events as periods of heightened public attention to sustainability. Section 5.2 examines alternative specifications for temperature and air pollution.

5.1 Google Searches

To provide external support for the public-attention interpretation, we use Google Trends data on sustainability-related search intensity in Singapore. The goal is to examine whether environment-related public events coincide with greater public attention to sustainability and whether such attention is associated with higher green delivery take-up.

Table 7 reports the results. Google search intensity offers a useful proxy for day-to-day variation in public attention to environmental issues. Columns 1 and 2 examine whether environment-related public events predict higher sustainability-related search activity. The estimates show that event days are associated with a sizable increase in the Google Trends sustainability index. This pattern suggests that the events we code correspond to periods when environmental issues become more visible to the public.

[Insert Table 7]

Columns 3 and 4 relate sustainability-related search intensity to green delivery choice. Higher search intensity is positively associated with green delivery take-up. A one-unit increase in the log sustainability index raises green choice by 0.03 percentage points without controls and by 0.06 percentage points with controls. Although these magnitudes are modest, the positive relationship is consistent with the view that green delivery adoption is higher when environmental issues receive greater public attention.

Overall, the Google Trends evidence supports the public-attention interpretation. Environment-related public events are associated with higher sustainability-related search activity, and periods of higher search activity are also associated with higher green delivery take-up. We interpret these results as complementary evidence consistent with temporary shifts in public attention being related to short-run changes in sustainable household behavior.

5.2 Alternative Environmental Measures

In what follows, we examine whether our main findings are sensitive to how environmental stress is measured. We consider two sets of robustness checks. First, we revisit the temperature results using log transformations and an alternative construction of abnormal temperature. Second, we examine whether the air-pollution results are robust to using PSI in levels rather than logs. These exercises address whether the baseline estimates depend on functional-form choices or on the specific construction of the environmental variables.

We begin with heat exposure. Table A5 in the Online Appendix re-estimates the effects of temperature using logarithmic transformations rather than the level measures

used in the main table. Under this specification, the coefficients can be interpreted as semi-elasticities. The results are consistent with the baseline findings. In the mean-temperature specification, a 10% increase is associated with about a 0.92 percentage point decline in green delivery adoption when delivery characteristics are included, with the estimate statistically significant at the 1% level. The maximum-temperature specification yields a similar negative relationship. The effect is larger for wet-bulb temperature: a 10% increase is associated with a 2.29 percentage point decline in green choice, also significant at the 1% level. This pattern is consistent with the level-temperature results and with wet-bulb temperature being a closer measure of experienced heat discomfort in Singapore.

Table A5 also reports results using an alternative abnormal temperature measure. This measure differs from the baseline in two ways. In the time dimension, the historical norm is computed by averaging daily maximum temperatures over a ± 5 -day window around each calendar date during 2011–2022, rather than using only the same calendar date. This smooths idiosyncratic fluctuations in the historical reference period. In the spatial dimension, district-level abnormal temperature is constructed using the four nearest weather stations rather than the two nearest stations, providing broader spatial coverage. The estimates are very close to the baseline results, indicating that the negative heat effect is not driven by a particular choice of temporal or spatial smoothing.

We next turn to air pollution. Table A6 in the online appendix re-estimates the pollution specification using the level of PSI rather than its logarithm. Under this specification, the coefficients measure the percentage-point change in green delivery adoption associated with a one-unit increase in the Pollutant Standards Index. The results again point in the same direction. In Column 1, a one-unit increase in PSI reduces green delivery adoption by about 0.03 percentage points, statistically significant at the 5% level. The estimate is virtually unchanged in Column 2 after adding delivery-menu and order controls, suggesting that the pollution effect is not explained by changes in the displayed menu or by order composition. In economic terms, a 10-point increase in PSI lowers green delivery take-up by approximately 0.3 percentage points. Moving from a relatively clean day with PSI around 25 to a moderately polluted day with PSI around 80 implies a decline of roughly 1.7 percentage points in green choice.

Taken together, these alternative measures support the baseline conclusion. Heat and air pollution are both negatively associated with green delivery adoption, and the pattern is robust to log temperature measures, alternative spatial smoothing, and alternative constructions of abnormal temperature. The main findings therefore do not appear to be artifacts of a specific temperature transformation or measurement choice.

6 Conclusion

This paper shows that attention scarcity can constrain household contributions to the green transition. Using transaction-level data from Singapore’s largest online grocery platform, we study adoption of a delivery option labeled Less CO₂, which is designed to facilitate route consolidation and reduce delivery-related emissions. We find that households are less likely to choose this option when temperatures rise and air quality deteriorates. By contrast, environment-related public events are associated with higher adoption, consistent with public environmental cues shifting attention toward environmental goals. Both heat and air pollution attenuate the event-day association, suggesting that environmental stress weakens the attentional channel through which public environmental cues may influence household behavior.

Our contribution is to show that environmental stress and public environmental cues jointly shape sustainable consumption, and that the event-day association is state-dependent. Environment-related public events are associated with a 5 to 7 percentage point increase in green delivery adoption on event days, with no visible pre-trends and rapid reversion afterward, consistent with a temporary attention shift. Heat exposure is associated with lower green choice: a one-degree Celsius increase in wet-bulb temperature lowers adoption by about 0.9 to 1.2 percentage points. Poorer air quality is also associated with lower green take-up, although the magnitude is smaller: a 10% increase in PSI lowers adoption by about 0.12 percentage points. Finally, the event-day association is weaker under both higher heat-humidity stress and worse air quality. A one-degree Celsius increase in wet-bulb temperature reduces the event-day association by about 3.4 percentage points, while a 10-point increase in PSI reduces it by about 3.5 percentage points.

The policy implication is that public environmental cues can shift behavior, but their effectiveness is conditional on cognitive bandwidth. As climate stress intensifies, mitigation policy may need to rely more on instruments that reduce cognitive demands or bypass active attention, including defaults, automation, and infrastructure that embeds sustainability into the choice set. The results highlight an overlooked cost of warming: rising temperatures can erode the effectiveness of behavioral mitigation tools precisely when they are most needed.

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Select delivery time

6-Hour Slot
\$5.99

2-Hour Slot
\$5.99 + \$3.99

Express
\$5.99 + \$5.99

Spend \$40.00 to get free shipping fee, saving \$5.99

Tue
25

Wed
26

Thu
27

Fri
28

Sat
29

Sun
30

☐ 2pm - 4pm

Less CO₂

☐ 3pm - 5pm

Less CO₂

☐ 4pm - 6pm

☐ 5pm - 7pm

☐ 6pm - 8pm

☐ 7pm - 9pm

☐ 8pm - 10pm

Figure 1: Screenshot of the App

Note. This figure displays the RedMart application user interface for selecting delivery time slots. Slots marked with a green ‘Less CO₂’ label indicate green delivery options, which consolidate delivery routes to reduce carbon emissions. The label is the only visible indicator identifying a slot as a green delivery option. The figure is illustrative; the estimation sample is restricted to checkout instances with zero slot fees.

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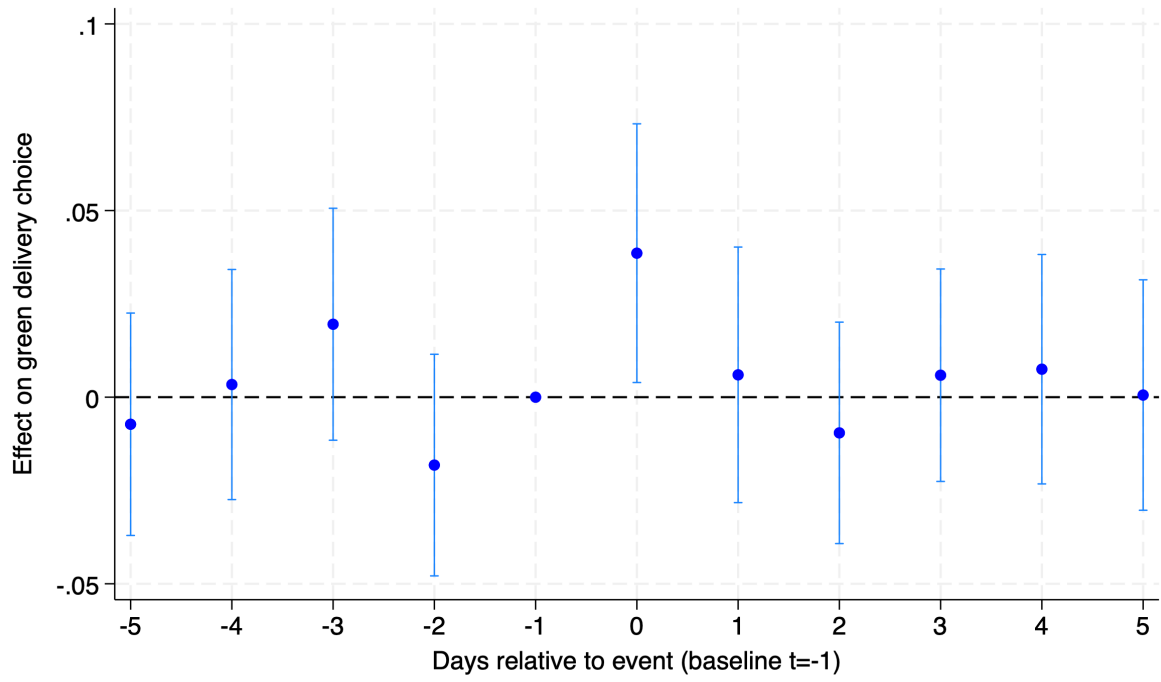


Figure 2: Event Study

Note. This figure plots the estimated coefficients and 90% confidence intervals from the event study specification in Equation (3). The dependent variable is green delivery choice. The omitted baseline category is $t=-1$ (one day before the event). The horizontal axis indicates days relative to the sustainability event, ranging from $t=-5$ to $t=5$. The vertical axis measures the change in green delivery adoption probability relative to the baseline. The absence of significant pre-trends ($t=-5$ to $t=-2$) supports the absence of differential pre-trends.

Table 1: Summary Statistics

Var.	Def.	Obs	Mean	SD	Min	Max
<i>Panel A: Delivery Characteristics</i>						
Green	1 if the customer selects a delivery slot labeled ‘Less CO ₂ ’, 0 otherwise.	325,555	0.408	0.491	0.000	1.000
Proportion	Green option percentage, = eco avail entries/avail entries	325,555	0.251	0.195	0.005	1.000
Expensive	1 if the order value is at least S\$100, 0 otherwise.	325,555	0.571	0.495	0.000	1.000
Avail entries	Number of available delivery time slots	325,555	15.830	20.611	1.000	260.000
Eco avail entries	Number of available green delivery time slots	325,555	2.250	2.318	1.000	81.000
Order weight	Weight of the order in metric tons	325,362	0.026	0.024	0.0002	1.283
Qty	Sum (quantity of each item)	325,555	20.857	16.132	1.000	499.000
<i>Panel B: Meteorological Data</i>						
Abn Temp	Daily abnormal temperature in degree Celsius, represented by the deviation from the historical maximum temperature	325,279	0.355	1.027	−4.360	3.242
Mean Temp	Mean of daily absolute temperature	325,555	28.292	1.221	23.900	30.700
Max Temp	Maximum of daily absolute temperature	325,555	31.970	1.623	25.300	35.800
Twb	District-level wet-bulb temperature	325,555	25.075	0.746	22.010	27.540
PSI	Mean of PSI, a number used to indicate the level of pollutants in air	325,555	45.418	11.931	14.292	108.375
<i>Panel C: Environment-related Public Event</i>						
HasEvent	1 if at least one environment-related public event takes place on that date, 0 otherwise.	325,555	0.087	0.282	0.000	1.000

Note. This table reports summary statistics for the estimation sample, which covers orders placed between March 2023 and March 2025 in Singapore. We restrict the sample to checkout instances with zero slot fees and with at least one delivery slot labeled ‘Less CO₂’ available to the customer, so that variation in green take-up is not mechanically driven by delivery-price differences or by the absence of a green option. Panel A presents the results of order and delivery characteristics, while Panel B summarizes environmental variables matched to the delivery district and date. It includes four measures of temperature (abnormal temperature Abn Temp, mean daily temperature, maximum daily temperature and wet-bulb temperature), as well as air pollution (PSI). Panel C reports the environment-related public-event indicator.

Table 2: The impact of heat on consumer green choice

	(1) Green	(2) Green
Panel A: Abnormal Temperature		
Abn Temp	-0.0043*** (0.0012)	-0.0036*** (0.0011)
Adj. R-squared	0.1351	0.1687
Observations	243,912	243,795
Panel B: Mean Temperature (levels)		
Mean Temp	-0.0042*** (0.0012)	-0.0032*** (0.0012)
Adj. R-squared	0.1352	0.1687
Observations	244,177	244,060
Panel C: Maximum Temperature (levels)		
Max Temp	-0.0026*** (0.0008)	-0.0024*** (0.0008)
Adj. R-squared	0.1352	0.1688
Observations	244,177	244,060
Panel D: Wet-bulb Temperature (levels)		
Twb	-0.0121*** (0.0023)	-0.0092*** (0.0022)
Adj. R-squared	0.1352	0.1688
Observations	244,177	244,060
Control	No	Yes
FE	Yes	Yes

Note. This table presents the impact of heat on consumer green choice. The dependent variable *Green* equals one if consumers choose the green delivery option. Panel A reports the effect of abnormal temperature: *Abn Temp* is daily abnormal temperature (in °C), defined as the deviation from the historical maximum temperature. Panels B–D report temperature measures in levels: *Mean Temp* and *Max Temp* are daily mean and maximum temperatures (in °C), and *Twb* is district-level wet-bulb temperature (in °C). Wet-bulb temperature is a proxy for human heat stress that combines air temperature and relative humidity; T_w is approximated from dry-bulb temperature T (in °C) and relative humidity RH (in %) following Stull (2011). Column (2) additionally controls for internal delivery variables *Proportion*, *Expensive*, *Qty*, and *Order weight*. All columns include individual fixed effects, slot timing fixed effects, district fixed effects, day-of-week fixed effects, and year-month fixed effects. Standard errors in parentheses are clustered at the individual level. ***, **, and * denote significance at the 1%, 5%, and 10% levels, respectively.

Table 3: The impact of air pollution on consumer green choice

	(1) Green	(2) Green
$\log(\text{PSI})$	-0.0117** (0.0047)	-0.0122*** (0.0043)
Control	No	Yes
FE	Yes	Yes
Adj. R-squared	0.1351	0.1404
Observations	244,177	244,060

Note. This table presents the main effect of air pollution on consumer green choice. The dependent variable *Green* equals one if the consumer chooses the green delivery option. $\log(\text{PSI})$ denotes the natural logarithm of the Pollutant Standards Index (PSI), Singapore's official air-quality index. Column (2) additionally controls for internal delivery variables *Proportion*, *Expensive*, *Qty*, and *Order weight*. All columns include individual fixed effects, slot timing fixed effects, district fixed effects, day-of-week fixed effects, and year-month fixed effects. Standard errors in parentheses are clustered at the individual level. ***, **, and * denote significance at the 1%, 5%, and 10% levels, respectively.

Table 4: The impact of environment-related public events on consumer green choice

	(1) Green	(2) Green
HasEvent	0.0703*** (0.0064)	0.0518*** (0.0063)
Control	No	Yes
FE	Yes	Yes
Observations	244,177	244,060
Adj. R-squared	0.1357	0.1690

Note. This table presents the main effect of environment-related public events on consumer green choice. The dependent variable *Green* equals one if the consumer chooses the green delivery option. *HasEvent* is a dummy variable equal to one when at least one environment-related public event is active (see Appendix Table A1 for the complete listing). Column (2) additionally controls for internal delivery variables *Proportion*, *Expensive*, *Qty*, and *Order weight*. All columns include individual fixed effects, slot timing fixed effects, district fixed effects, day-of-week fixed effects, and year-month fixed effects. Standard errors are in parentheses. ***, **, and * denote significance at the 1%, 5%, and 10% levels, respectively.

Table 5: Interaction between Environment-Related Public Events and Environmental Stressors

	(1) Green	(2) Green
Twb	-0.0044* (0.0023)	
PSI		0.0003** (0.0001)
HasEvent x Twb	-0.0341*** (0.0055)	
HasEvent x PSI		-0.0035*** (0.0003)
HasEvent	0.9017*** (0.1372)	0.2135*** (0.0171)
Control	Yes	Yes
FE	Yes	Yes
Adj. R-squared	0.1692	0.1695
Observations	244,060	244,060

Note. This table presents the interaction between environment-related public events and environmental stressors on consumer green choice. The dependent variable *Green* equals one if the consumer chooses the green delivery option. *HasEvent* equals one if any environment-related public event takes place on that date, and zero otherwise. *Twb* denotes district-level wet-bulb temperature in degrees Celsius, and *PSI* denotes the Pollutant Standards Index. Both columns additionally control for internal delivery variables *Proportion*, *Expensive*, *Qty*, and *Order weight*. All columns include individual fixed effects, slot timing fixed effects, district fixed effects, day-of-week fixed effects, and year-month fixed effects. Standard errors in parentheses are clustered at the individual level. ***, **, and * denote significance at the 1%, 5%, and 10% levels, respectively.

Table 6: Drivers of Green Delivery (Order level)

	(1)	(2)	(3)	(4)
	Green	Green	Green	Green
Proportion	0.4798*** (0.0062)			
Expensive		-0.0742*** (0.0030)		
Qty			-0.0029*** (0.0001)	
Order weight				-0.8853*** (0.0731)
FE	Yes	Yes	Yes	Yes
Adj. R-squared	0.1632	0.1382	0.1390	0.1358
Observations	244,177	244,177	244,177	244,060

Notes. This table presents the drivers of green delivery based on order-level data. The dependent variable *Green* is whether consumers choose the green option. *Proportion* is the percentage of green option in all options. *Expensive* is measured by the total order price, if it is greater than or equal to 100, it equals 1, and 0 otherwise. *Qty* is the sum quantity of each item. *Order weight* is the weight of the order in metric tons. All the columns control for individual fixed effects, slot timing fixed effects, district fixed effects, day-of-week fixed effects and year-month fixed effects. Column (1) represents the effect of the share of green options on consumer green choice. Column (2) represents the effect of order price on consumer green choice. Column (3) represents the effect of item quantity on consumer green choice. Column (4) explains the effect of order weight on consumer green choice. Standard errors in parentheses are clustered at the individual level. ***, **, and * denote significance at the 1%, 5%, and 10% levels, respectively.

Table 7: The effect of Google Trends on consumer green choice

	(1) Sustainability	(2) Sustainability	(3) Green	(4) Green
HasEvent	1.5675*** (0.1480)	1.6528*** (0.1483)		
Sustainability			0.0003** (0.0001)	0.0006*** (0.0001)
Control	No	Yes	No	Yes
FE	Yes	Yes	Yes	Yes
Adj. R-squared	0.6820	0.6826	0.1217	0.1700
Observations	226,323	226,214	226,323	226,214

Note. The table presents the effect of environment-related Google Trends index on consumer green choice. In Column (1)-(2), the dependent variable is *Sustainability*, which denotes the natural logarithm of the Google Trends index of sustainability. In Column (3)-(4), the dependent variable is *Green*, which represents whether consumers choose green option. Column (1) and (3) do not include controls while Columns (2), (4) control for internal delivery variables *Proportion*, *Expensive*, *Qty*, *Order weight*. All columns include individual fixed effects, slot timing fixed effects, district fixed effects, day-of-week fixed effects and year-month fixed effects. Standard errors in parentheses are clustered at the individual level. ***, **, and * denote significance at the 1%, 5%, and 10% levels, respectively.

Internet Appendix

Not Intended for Publication

A Additional Tables and Figures

Table A1: Listing of Events

Start Date	End Date	EventName	Description	Duration Days
2023/5/13	2023/5/13	SG Clean Day - Islandwide Clean Up	Record-breaking community clean-up under Public Hygiene Council	1
2023/9/14	2023/9/14	Recycle Right Campaign Update 2023	NEA update encouraging correct recycling; household recycling survey release	1
2023/9/16	2023/9/16	World Cleanup Day SG 2023	Global cleanup day observed across Singapore	1
2023/10/14	2023/10/14	International E-waste Day activation (NEA)	NEA e-waste awareness activation at Toa Payoh HDB Hub	1
2023/11/4	2023/11/4	Clean & Green Singapore Day 2023	National CGS Day with community activities and tree-planting	1
2023/12/15	2023/12/15	GreenGov.SG Report (Inaugural)	Public sector sustainability performance report released	1
2024/3/4	2024/3/4	Green Plan 2030 - COS Update 2024	Committee of Supply speech / Green Plan progress highlights	1
2024/6/12	2024/7/14	Go Green SG 2024	Nationwide sustainability movement; many public events	33
2024/6/14	2024/7/22	Singapore Oil Spill Response	Collision-led fuel oil spill; response and phased beach reopening	39
2024/6/24	2024/6/24	NEA Update: Bulk Oil Removal Near Completion	NEA update on oil removal progress; shift to specialised cleanup	1
2024/7/13	2024/7/30	Dry Spell (18 days)	18 consecutive days with <1.0 mm rainfall; respite end July	18
2024/9/20	2024/9/20	World Cleanup Day SG 2024	Global cleanup day	1
2024/11/3	2024/11/3	Clean & Green Singapore Day 2024	Environmental stewardship day; community and hygiene focus	1
2024/11/18	2024/11/18	GreenGov.SG Report FY2023	Annual public sector sustainability performance report release	1
2025/1/10	2025/1/13	Northeast Monsoon Surge Advisory Window	NEA advisory: monsoon surge bringing cool rainy conditions	4

Notes. This table lists the environment-related public events used to construct our event-day indicator. Events are hand-collected from official announcements and updates by the Ministry of Sustainability and the Environment (MSE), the National Environment Agency (NEA), and affiliated public programs, and are coded with a start date and end date. *Duration Days* is computed as (end date – start date + 1), so one-day events have duration equal to 1. In the main analysis, we define an event day as any calendar day that falls within the start–end window of an event listed here. When multiple events overlap in time, the event-day indicator remains equal to one. In the event-study analysis, we group consecutive event days into a single event episode and drop days that fall into overlapping pre/post windows to avoid contamination (see Appendix C for details).

Table A2: Interaction between Environment-Related Public Events and Environmental Stressors: No-Control Specifications

	(1) Green	(2) Green	(3) Green	(4) Green	(5) Green
Abn Temp	0.0004 (0.0012)				
Mean Temp		-0.0010 (0.0013)			
Max Temp			0.0001 (0.0009)		
Twb				-0.0058** (0.0023)	
PSI					0.0002* (0.0001)
HasEvent x Abn Temp	-0.0364*** (0.0035)				
HasEvent x Mean Temp		-0.0160*** (0.0029)			
HasEvent x Max Temp			-0.0133*** (0.0021)		
HasEvent x Twb				-0.0433*** (0.0056)	
HasEvent x PSI					-0.0044*** (0.0004)
HasEvent	0.0742*** (0.0065)	0.5170*** (0.0813)	0.4827*** (0.0665)	1.1498*** (0.1412)	0.2716*** (0.0176)
Control	No	No	No	No	No
FE	Yes	Yes	Yes	Yes	Yes
Adj. R-squared	0.1362	0.1359	0.1359	0.1361	0.1364
Observations	243,912	244,177	244,177	244,177	244,177

Note. This table presents the interaction between environment-related public events and environmental stressors on consumer green choice without internal delivery controls. The dependent variable *Green* equals one if the consumer chooses the green delivery option. *HasEvent* equals one if any environment-related public event takes place on that date, and zero otherwise. *Abn Temp* is daily abnormal temperature in degrees Celsius. *Mean Temp*, *Max Temp*, and *Twb* are daily mean temperature, daily maximum temperature, and district-level wet-bulb temperature in degrees Celsius. *PSI* denotes the Pollutant Standards Index. All columns include individual fixed effects, slot timing fixed effects, district fixed effects, day-of-week fixed effects, and year-month fixed effects. Standard errors in parentheses are clustered at the individual level. ***, **, and * denote significance at the 1%, 5%, and 10% levels, respectively.

Table A3: Interaction between Environment-Related Public Events and Environmental Stressors: Additional Temperature Specifications with Controls

	(1) Green	(2) Green	(3) Green
Abn Temp	0.0002 (0.0012)		
Mean Temp		-0.0006 (0.0013)	
Max Temp			-0.0003 (0.0008)
HasEvent x Abn Temp	-0.0296*** (0.0034)		
HasEvent x Mean Temp		-0.0137*** (0.0028)	
HasEvent x Max Temp			-0.0110*** (0.0021)
HasEvent	0.0554*** (0.0064)	0.4337*** (0.0793)	0.3901*** (0.0646)
Control	Yes	Yes	Yes
FE	Yes	Yes	Yes
Adj. R-squared	0.1693	0.1691	0.1692
Observations	243,795	244,060	244,060

Note. This table presents the interaction between environment-related public events and temperature-based environmental stressors on consumer green choice with internal delivery controls. The dependent variable *Green* equals one if the consumer chooses the green delivery option. *HasEvent* equals one if any environment-related public event takes place on that date, and zero otherwise. *Abn Temp* is daily abnormal temperature in degrees Celsius. *Mean Temp* and *Max Temp* are daily mean and maximum temperatures in degrees Celsius. All columns additionally control for internal delivery variables *Proportion*, *Expensive*, *Qty*, and *Order weight*. All columns include individual fixed effects, slot timing fixed effects, district fixed effects, day-of-week fixed effects, and year-month fixed effects. Standard errors in parentheses are clustered at the individual level. ***, **, and * denote significance at the 1%, 5%, and 10% levels, respectively.

Table A4: Interaction between Environment-Related Public Events and Environmental Stressors: Log Specifications with Controls

	(1) Green	(2) Green	(3) Green	(4) Green
log(Mean Temp)	-0.0184 (0.0354)			
log(Max Temp)		-0.0102 (0.0261)		
log(Twb)			-0.1081* (0.0571)	
log(PSI)				0.0080* (0.0048)
HasEvent x log(Mean Temp)	-0.3664*** (0.0781)			
HasEvent x log(Max Temp)		-0.3184*** (0.0626)		
HasEvent x log(Twb)			-0.8403*** (0.1361)	
HasEvent x log(PSI)				-0.1539*** (0.0155)
HasEvent	1.2698*** (0.2598)	1.1425*** (0.2148)	2.7532*** (0.4380)	0.6327*** (0.0587)
Control	Yes	Yes	Yes	Yes
FE	Yes	Yes	Yes	Yes
Adj. R-squared	0.1691	0.1691	0.1692	0.1695
Observations	244,060	244,060	244,060	244,060

Note. This table reports log specifications corresponding to the interaction between environment-related public events and environmental stressors on consumer green choice. The dependent variable *Green* equals one if the consumer chooses the green delivery option. *HasEvent* equals one if any environment-related public event takes place on that date, and zero otherwise. *log(Mean Temp)*, *log(Max Temp)*, *log(Twb)*, and *log(PSI)* denote the natural logarithms of mean temperature, maximum temperature, wet-bulb temperature, and the Pollutant Standards Index, respectively. All columns additionally control for internal delivery variables *Proportion*, *Expensive*, *Qty*, and *Order weight*. All columns include individual fixed effects, slot timing fixed effects, district fixed effects, day-of-week fixed effects, and year-month fixed effects. Standard errors in parentheses are clustered at the individual level. ***, **, and * denote significance at the 1%, 5%, and 10% levels, respectively.

Table A5: Heat (Robustness: Alternative Abnormal Temperature and Temperature in Logs)

	(1) Green	(2) Green
Panel A: Abnormal Temperature		
Abn Temp 4stn	-0.0043*** (0.0012)	-0.0035*** (0.0012)
Adj. R-squared	0.1351	0.1687
Observations	243,912	243,795
Panel B: Mean Temperature (log)		
log(Mean Temp)	-0.1184*** (0.0339)	-0.0919*** (0.0333)
Adj. R-squared	0.1352	0.1687
Observations	244,177	244,060
Panel C: Maximum Temperature (log)		
log(Max Temp)	-0.0830*** (0.0251)	-0.0765*** (0.0246)
Adj. R-squared	0.1352	0.1688
Observations	244,177	244,060
Panel D: Wet-bulb Temperature (log)		
log(Twb)	-0.2996*** (0.0565)	-0.2294*** (0.0552)
Adj. R-squared	0.1352	0.1688
Observations	244,177	244,060
Control	No	Yes
FE	Yes	Yes

Note. This table reports robustness results using an alternative abnormal-temperature construction and log-transformed temperature measures. Panel A uses *Abn Temp 4stn*, measured as the purchase-day maximum temperature minus the historical average of the maximum temperature for the corresponding date and its ± 5 -day window over 2011-2022. The district-level abnormal temperature is constructed by averaging the nearest four weather stations and is in $^{\circ}\text{C}$. Panels B–D use log-transformed temperature measures: $\log(\text{Mean Temp})$ is the natural logarithm of daily mean temperature, $\log(\text{Max Temp})$ is the natural logarithm of daily maximum temperature, and $\log(\text{Twb})$ is the natural logarithm of district-level wet-bulb temperature. Column (2) additionally controls for internal delivery variables *Proportion*, *Expensive*, *Qty*, and *Order weight*. All columns include individual, slot timing, district, day-of-week, and year-month fixed effects. Standard errors in parentheses are clustered at the individual level. ***, **, and * denote significance at the 1%, 5%, and 10% levels, respectively.

Table A6: Robustness: Air Pollution (Levels)

	(1) Green	(2) Green
PSI_mean	-0.0003** (0.0001)	-0.0003** (0.0001)
Control	No	Yes
FE	Yes	Yes
Observations	244,177	244,060
R-squared	0.3544	0.3583
Adj. R-squared	0.1351	0.1403

Note. This table presents the impact of air pollution on consumer green choice and reports robustness results for air pollution using PSI in levels rather than logs. *PSI_mean* denotes the daily Pollutant Standards Index matched to the delivery location. Column (2) additionally controls for internal delivery variables *Proportion*, *Expensive*, *Qty*, and *Order weight*. All columns include individual, slot timing, district, day-of-week, and year-month fixed effects. Standard errors in parentheses are clustered at the individual level. ***, **, and * denote significance at the 1%, 5%, and 10% levels, respectively.

B Spatial and Temporal Variation in Wet-Bulb Temperature across Singapore

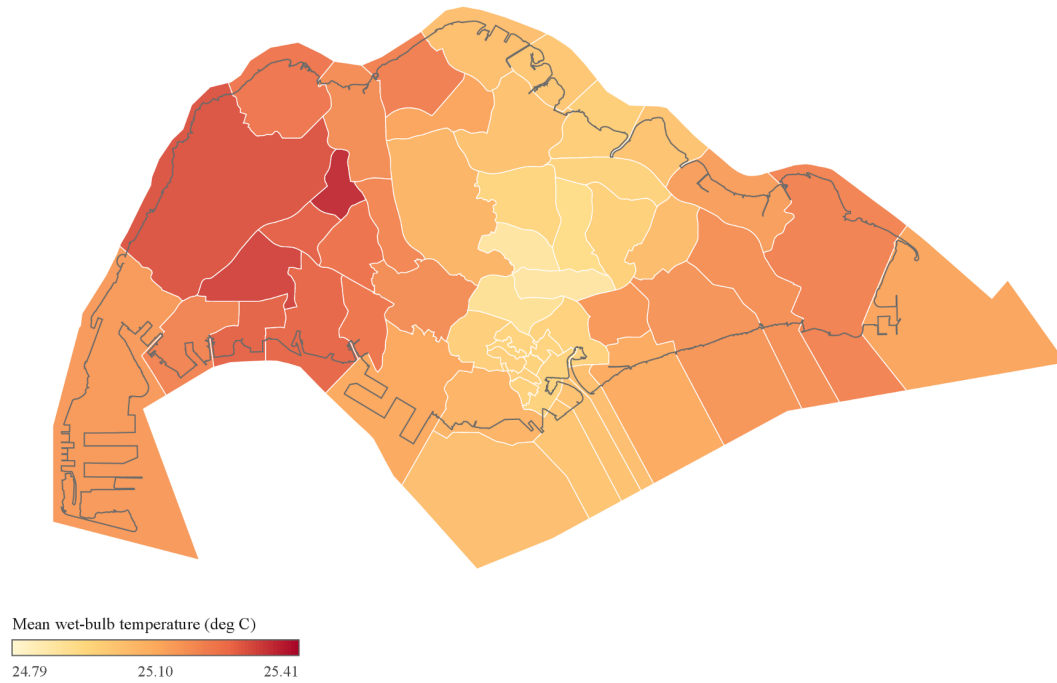


Figure B1: Mean District Wet-Bulb Temperature across Singapore

Notes: The sample covers March 1, 2023 to April 1, 2025. The figure visualizes mean district wet-bulb temperature using the SLA National Map mainland coastline and planning-area polygons as the display surface. Color is interpolated from the 28 postal-district centroids because official postal-district polygons are not available. Offshore Western Islands and North-Eastern Islands are omitted for readability.

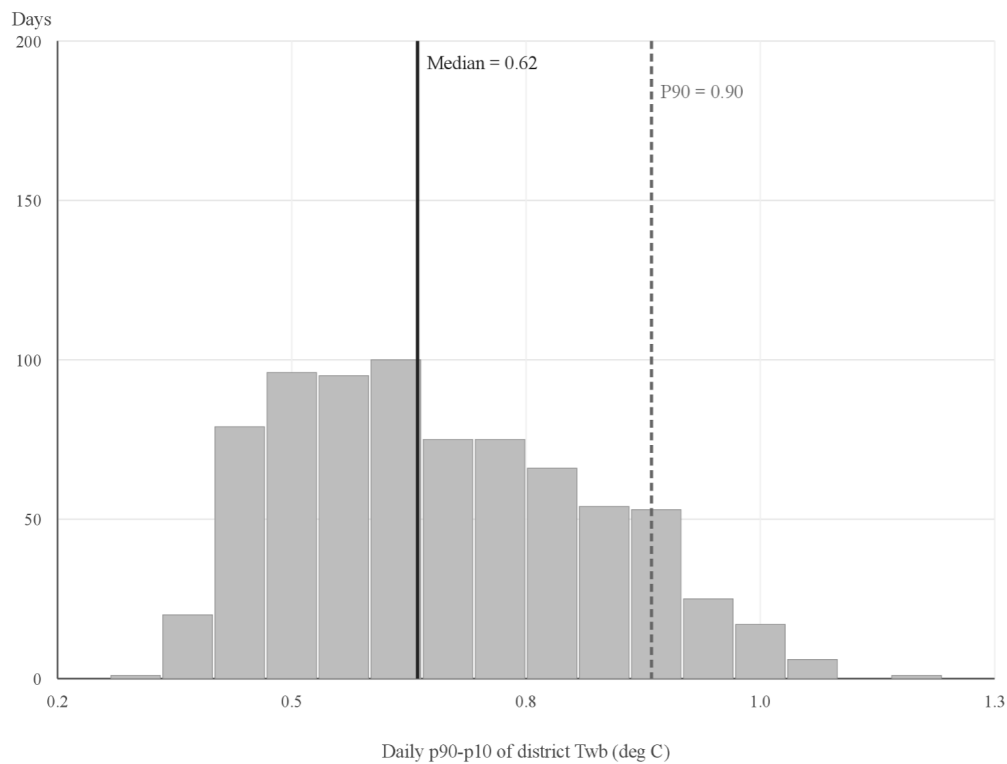


Figure B2: Distribution of Daily Cross-District Variation in Wet-Bulb Temperature
Notes: The sample covers March 1, 2023 to April 1, 2025. The figure plots the distribution of the daily p90-p10 in district wet-bulb temperature across the 28 postal districts.

C Event Study Construction and Overlap Dropping

This appendix describes how we construct relative-time indicators around event episodes and how we drop overlapping windows.

C.1 Event episodes

Let E_t be an indicator equal to one on calendar date t if an environment-related public event is active. An event episode is a maximal consecutive sequence of dates with $E_t = 1$. We define an episode start as a date with $E_t = 1$ and $E_{t-1} = 0$, and an episode end as a date with $E_t = 1$ and $E_{t+1} = 0$.

C.2 Relative-time indicators

For each episode, we define pre-period indicators for $k = 1, \dots, K$ as dates that are exactly k days before an episode start and are not themselves event days. Similarly, we define post-period indicators for $k = 1, \dots, K$ as dates that are exactly k days after an episode end and are not event days. We set $K = 14$.

We then define event time $\tau(t)$ as follows: (i) $\tau(t) = 0$ if $E_t = 1$ (event day within an episode); (ii) $\tau(t) = -k$ if t is the k th day before an episode start; and (iii) $\tau(t) = +k$ if t is the k th day after an episode end. The omitted category in the event study is $\tau = -1$.

C.3 Dropping overlapping windows

When episodes occur close in time, a calendar date can fall within the pre-window of one episode and the post-window of another. To avoid such contamination, we drop any date t that satisfies both conditions: t lies in the K -day pre-window of some episode and in the K -day post-window of some (other) episode. After dropping these overlapping dates, each remaining date in the event-study sample belongs to at most one episode window.

D Proofs for Section 4

Proof of Proposition 1. By the chain rule,

$$\frac{\partial P(G)}{\partial \tilde{S}} = \Phi_m \cdot \frac{\partial m}{\partial \tilde{S}}.$$

Equation (16) implies $\Phi_m > 0$, and Equation (12) implies $\partial m / \partial \tilde{S} < 0$ for $E \in \{0, 1\}$. Therefore $\partial P(G) / \partial \tilde{S} < 0$. $\square$

Proof of Proposition 2. From Equation (11),

$$m_1 \equiv m(\tilde{S}, 1) = \bar{m} + \sigma - (\alpha + \beta)\tilde{S}, \quad m_0 \equiv m(\tilde{S}, 0) = \bar{m} - \alpha\tilde{S}.$$

Hence

$$m_1 - m_0 = \sigma - \beta\tilde{S},$$

which is strictly positive on the observed support under Event-Attention Condition E in Equation (14). Since $\Phi(m, \phi)$ is strictly increasing in m , we have $\Phi(m_1, \phi) > \Phi(m_0, \phi)$, i.e.

$$P(G \mid E = 1, \tilde{S}) - P(G \mid E = 0, \tilde{S}) > 0.$$

$\square$

Proof of Proposition 3. Let

$$\Delta_E P(G; \tilde{S}) = \Phi(m_1, \phi) - \Phi(m_0, \phi),$$

where

$$m_1 = \bar{m} + \sigma - (\alpha + \beta)\tilde{S}, \quad m_0 = \bar{m} - \alpha\tilde{S}.$$

Differentiating with respect to $\tilde{S}$,

$$\begin{aligned} \frac{d\Delta_E P(G; \tilde{S})}{d\tilde{S}} &= \Phi_m(m_1, \phi) (-\alpha - \beta) - \Phi_m(m_0, \phi) (-\alpha) \\ &= \alpha [\Phi_m(m_0, \phi) - \Phi_m(m_1, \phi)] - \beta \Phi_m(m_1, \phi). \end{aligned}$$

The second term is strictly negative because $\beta > 0$ and $\Phi_m > 0$. Under Event-Attention Condition E, $m_1 > m_0$. At states where both event and non-event adoption lie below one-half, $\Phi_{mm} > 0$, so Φ_m is increasing in m ; therefore $\Phi_m(m_1, \phi) \geq \Phi_m(m_0, \phi)$ and the first term is weakly negative. Hence

$$\frac{d\Delta_E P(G; \tilde{S})}{d\tilde{S}} < 0.$$

□

Proof of Proposition 4. Equation (18) implies

$$\frac{\partial P(G)}{\partial \phi} = \gamma \Phi(1 - \Phi) > 0$$

whenever $\gamma > 0$.

□